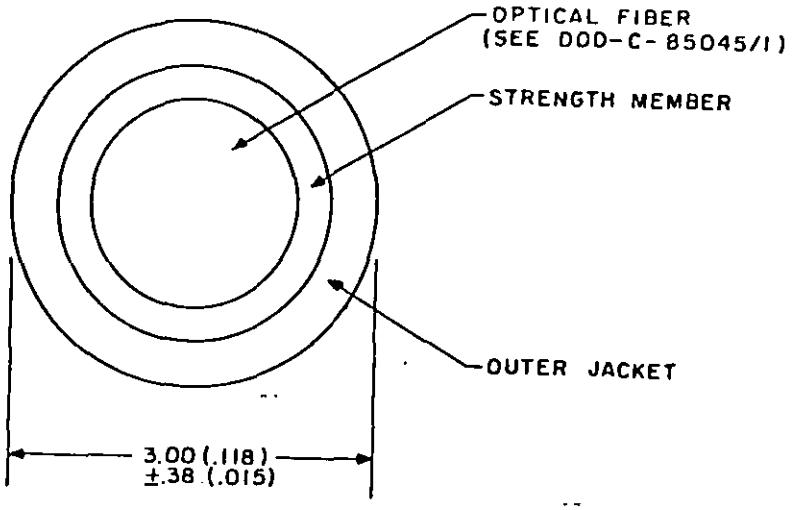
MILITARY SPECIFICATION SHEET

CABLES, FIBER OPTICS, BREAK OUT, INDIVIDUALLY JACKETED FIBERS
(METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cables described herein shall consist of this specification and the latest issue of DOD-C-85045.



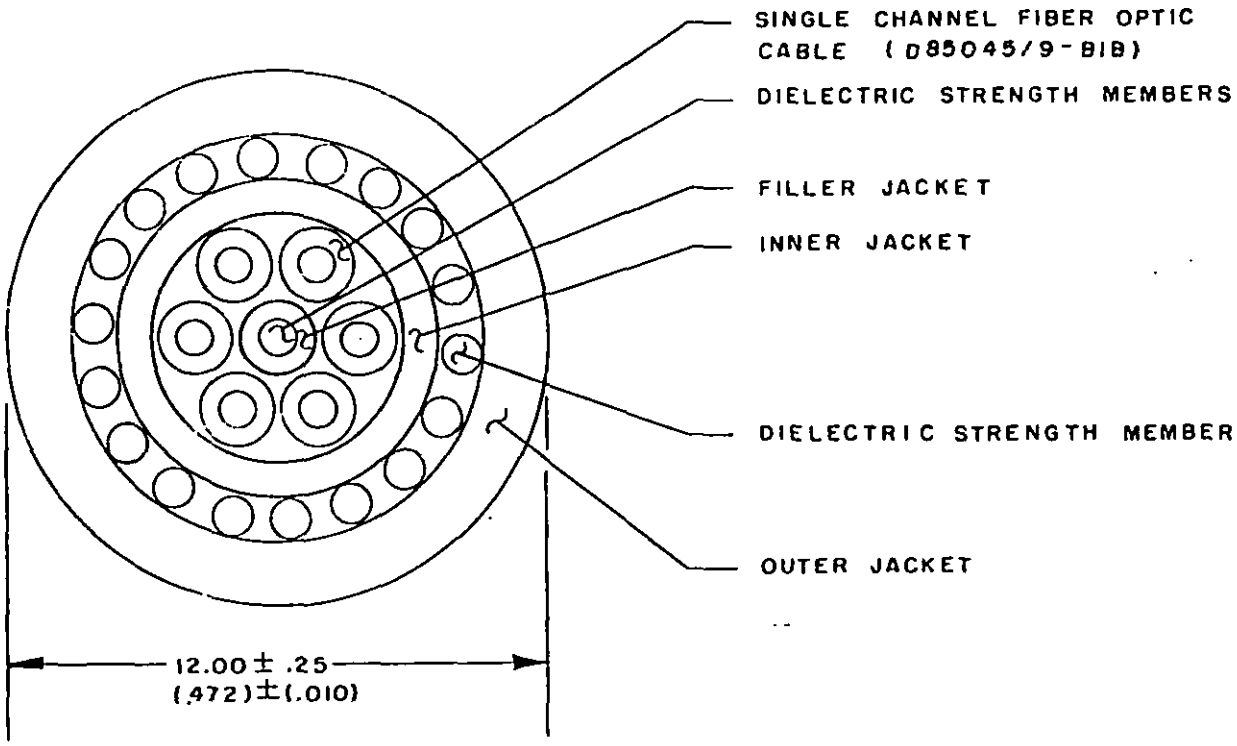
NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Inches are in parentheses.

FIGURE 1. One fiber cable.

AMSC N/A FSC 6015

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NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Inches are in parentheses.

FIGURE 2. Six fiber cable.

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TABLE I. Part numbers.

Part number D85045/9-	Number of fibers	Maximum weight (kg/km)
B1A	1	8
B1B	1	8
C1A	1	8
C1B	1	8
C6B	6	125

REQUIREMENTS:

Dimensions and configuration: Fiber cables shall be designed as in figures 1 and 2.

Weight: See table I.

Fiber: Core/cladding diameter and fiber type per part number designation.

Number of fibers: See table I.

Jacket material: Polyurethane or equivalent.

Filler and strength member material: Aramid yarn or equivalent.

Color: Outer cable jacket color shall be black.

Attenuation: Attenuation shall not exceed 1.5 dB/km at 1100 to 1300 nm for each and every fiber in the cable.

Optical fiber bandwidth The optical fiber bandwidth shall meet or exceed 800 MHz-km at 1300 ±20 nm.

Numerical aperture: 0.20 ±0.02.

Low temperature flexibility (cold bend): Not applicable.

Tight tube buffers: Constructed of thermoplastic polymer.

Crush: Applicable test shall be conducted at +25°C ±2°C. The test load of 500 newtons shall be applied within 5 seconds; hold time shall be 10 seconds minimum, release time shall be 5 seconds maximum.

Cable twist - bend: Applicable 100 cycles.

Tensile loading: 300 newtons minimum.

Storage temperature: The cable shall be stored for 240 hours minimum at -30°C ±2°C and +60°C ±2°C.

Fluid immersion (DOD-STD-1678, method 8030): After testing, the cable diameter shall not exceed the maximum cable diameter specified. Fluids, temperature and immersion time shall be as specified below.

Fluid	Temperature (C°)	Immersion time (hours)
Lubricating Oil MIL-L-23699	48 - 50	20
Automobile Gas MIL-G-3056	20 - 25	168

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Flammability: 60° angle procedure.

Jacket opacity: Shall be verified at 850 nm ±50 nm and at 1300 nm ±50 nm.

Parameter applicability: See table II.

Gel filling and flooding compound compatibility: The gel filling and flooding compound shall be compatible with the fiber buffer material.

TABLE II. Parameter applicability. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Group I					
Visual and mechanical	6 units 1 km each	X	X		
Attenuation rate	6 units 1 km each	X	X		
Optical fiber bandwidth	6 units 1 km each	X	X		
Crosstalk	6 units 1 km each	X	X		
Numerical aperture	6 units 2 m each (1 each from 6 cables)	X	X		
Group II					
Humidity	2 units 1 km each	X		X	
Storage temperature	1 units 1 km each (1 on reel, 1 off reel)	X		X	
Group III (Military environments)					
Crush	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Cable twist-bending	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Jacket self-adhesion or blocking	2 units >5 m each (1 each from 2 cables)	X			X
Cable jacket opacity	3 units 2 m each (1 each from 2 cables)	X			X
Fluid immersion	2 units (for each test fluid required) 2 m each (1 each from 2 cables)	X			X

See footnotes at end of table.

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TABLE II. Parameter applicability - Continued. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Cable element removability	3 units 2 m each (1 each from 3 cables)	X			X
Durability of identification marking	3 units 2 m each (1 each from 3 cables)	X			X
Flammability	3 units .6 m each (1 each from 3 cables)	X			X

1/ Tolerance on 1 km lengths is ±2% provided results are normalized to 1 km.

2/ Tolerance on shorter lengths is ±10%.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, SC
Navy - AS
Air Force - 11
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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